

Module 1

Explain topology and different types of topology
Illustrate and explain the different type of delays in network
Explain the working of circuit switching
Explain the working of packet switching
Explain in detail OSI model
Explain in detail TCP/IP model
Explain different techniques to improve QoS
Explain congestion control techniques in detail.

Module 2

Differentiate HTTP request and response
Explain working of FTP
Explain the working of SMTP
Explain the working of DNS and describe different types of records
Differentiate POP3 and IMAP
Explain pull protocols with example
Explain push protocols with example
Differentiate persistent and non persistent

Module 3

Explain connection oriented and connectionless multiplexing and demultiplexing
Explain the different services provided by transport layer
Differentiate selective repeat and go back N
Study rdt in detail with diagrams
Describe the structure of UDP segment and explain advantages of using UDP over TCP
Explain TCP segment structure
Explain TCP congestion control
TCP three way handshake for connection establishment with states.
Explain the principles of congestion control with fairness.

Module 4

Explain working of DHCP
Explain working virtual circuits
Explain working datagram networks
Differentiate IPV4 and IPV6
Explain working of distance vector algorithms
Explain working of link state algorithms
Explain working of autonomous system and its advantages .
Explain routing algorithms and different types of routing algorithms (link state and distance brief required)
Explain working of OSPF
Explain working of BGP
Explain working of RIP

Module 5

Explain different error detection techniques with example
Explain channel partitioning protocol and its types
Differentiate pure aloha and slotted aloha
Compare CSMA and CSMA/CD
Explain random access protocol in detail
Explain network management architecture in detail
Explain wifi architecture with neat diagram
Explain 802.11 MAC protocol.
Explain working of ethernet frame.

Describe firewall and its types
Describe VPN and its types.
Explain types of threats and attacks.
Explain SNMP working and its components